

Master's Thesis Report

Railway Flood-Risk Digital Twin for the SNCF Tartaiguille Corridor

Validation, Architecture & Scientific Calibration

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This report presents the design, implementation, and validation of a 4-layer railway flood-risk Digital Twin for the SNCF Tartaiguille corridor (Ligne 400, Drôme). The system integrates real-time meteorological data (Open-Meteo API), a hydrological screening layer based on the Soil Water Index (SWI), 2D hydraulic modeling (HEC-RAS), and vulnerability assessment using log-normal fragility curves calibrated against field data from Tsubaki et al. (2016).

The architecture implements a "Funnel Strategy" where the lightweight SWI layer screens the corridor in sub-second time, triggering the computationally expensive HEC-RAS 2D engine only when soil saturation exceeds a critical threshold ($SWI > 100$ mm). The F5 Grouped Architecture partitions the corridor into 21 sections, consolidating 107 physical assets (track segments, embankments, culverts, and bridges) to evaluate safety thresholds. The system resolves raw grid-level WSE outputs from pre-computed HEC-RAS HDF5 files (Plan 2: 21092025 Cévenol Storm) to generate RAMS-compliant alert levels (GREEN/YELLOW/ORANGE/RED) using a worst-case group roll-up.

A sensitivity analysis of the SWI half-life parameter ($T = 3\text{--}60$ days) demonstrates model robustness, and a historical Cévenol storm replay validates end-to-end pipeline functionality. This report details the final twin architecture, documents database inventory schemas, addresses map-rendering heuristics (depth-based alpha masking), and outlines a maturity roadmap for future live data assimilation.

Keywords: Digital Twin, Railway Infrastructure, Flood Risk, HEC-RAS, SWI, Fragility Curves, F5 Grouping, SNCF, RAMS

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1. Introduction

1.1 Problem Statement

Railway infrastructure in southern France is increasingly exposed to flash-flood hazards driven by intense Cévenol storm events. The Tartaiguille corridor (Ligne 400, PK 610–660) crosses the Drôme valley, where Mediterranean precipitation can generate rapid surface runoff threatening track stability. SNCF's RISK-VIP program (Cheetham et al., 2016) established that conventional periodic inspection cannot capture the temporal dynamics of flood exposure, necessitating real-time monitoring solutions.

1.2 Objectives

This thesis develops a Digital Twin prototype that:

1. **Screens** the corridor in real-time using a Soil Water Index (SWI) to identify saturation conditions.
2. **Simulates** flood hydraulics via HEC-RAS 2D when saturation thresholds are exceeded.
3. **Groups** assets functionally to align hydraulic water surface elevations with structural dependencies (F5 Grouping).
4. **Evaluates** structural vulnerability using fragility curves calibrated against published field data.
5. **Dispatches** RAMS-compliant alerts (speed restriction/halt) to operations control.

1.3 Scope

The system is designed for the 50 km Tartaiguille section with:

- **DTM resolution:** 1m (LiDAR-derived, Terrain.lhd_fx_lasd1.tif)
- **Temporal resolution:** 15-minute operational cycle
- **Input:** Open-Meteo API (historical + 48h forecast)
- **Output:** RAMS traffic-light alerts per BIM asset and section group
- **Database:** 107 assets mapped across 21 corridor sections

2. Literature Review

2.1 Digital Twins in Infrastructure

The concept of a Digital Twin (DT) for infrastructure management has evolved from static Building Information Models (BIM) toward dynamic, data-driven systems. Kaewunruen et al. (2021) demonstrated a BIM-embedded digital twin for the Taipei MRT, integrating lifecycle assessment with 3D visualization, but their system lacked real-time sensor fusion. Kim et al. (2025) advanced the state-of-the-art with a stormwater digital twin for Austin's Waller Creek watershed, achieving KGE improvement from 0.633 to 0.786 and sensor fault detection through Extended Kalman Filtering (EKF).

Pedersen (2023) defines three maturity levels:

- **Digital Model:** offline simulation, no live data.
- **Digital Shadow:** unidirectional data flow (physical → digital).
- **Digital Twin:** bidirectional data flow with feedback loop.

Table 1: Comparison of Digital Twin implementations in infrastructure management.

Reference	Domain	DT Maturity	Data Assimilation	Validation
Kaewunruen et al. (2021)	Railway MRT	Digital Model	None	BIM quantities
Kim et al. (2025)	Stormwater	Digital Twin	EKF	KGE=0.786, AUC=0.99

Cheetham et al. (2016)	Railway flood	Digital Shadow	None	87.5% incident capture
This work	Railway flood	Digital Shadow	None (future)	See Section 5

2.2 Soil Water Index (SWI) for Flood Screening

The Soil Water Index quantifies antecedent soil moisture as a precursor to surface runoff. Siva Subramanian et al. (2025) implemented SWI-based thresholds for Japan's Te-LEWS landslide early warning system. The SWI Leaky Bucket model used in this work follows the recursive formulation:

$$SWI(t) = R(t) + SWI(t-1) \times C$$

Where $R(t)$ is rainfall intensity (mm/h), $C = 2^{-1/(T \times 24)}$ is the hourly decay constant, and T is the half-life parameter (days).

2.3 HEC-RAS 2D for Railway Flood Modeling

The Hydrologic Engineering Center's River Analysis System (HEC-RAS) solves the 2D shallow water equations on unstructured grids. HEC-RAS is preferred for overland flood modeling due to its subgrid bathymetry, terrain-following mesh capabilities, and integration with LiDAR-derived DTMs.

2.4 Fragility Curves for Railway Vulnerability

Tsubaki et al. (2016) developed the first fragility curves specifically for railway embankment and ballast scour, using data from two Japanese flood events:

- **Ballast scour:** median $\Delta h = 0.30 \text{ m}$, $\sigma = 0.035$ (normal), $n = 15$ observations.
- **Combined (ballast + embankment):** median $\Delta h = 0.22 \text{ m}$, $\sigma = 0.15$ (log-normal), $n = 31$ observations.

These curves were validated against aerial photograph flood extents and are adapted here to define structural vulnerability.

3. Study Area & Data

3.1 Tartaiguille Corridor

The study area covers the Tartaiguille section of SNCF Ligne 400 (Montélimar–Marseille) in the Drôme department (44.65°N, 4.91°E). The corridor traverses a valley susceptible to flash flooding during Cévenol storm events, which can deliver 200+ mm of precipitation within 24 hours.

3.2 Digital Terrain Model

Table 2: DTM characteristics.

Parameter	Value
File	Terrain.lhd_fx_lasd1.tif
Resolution	~1 m (LiDAR)
CRS	EPSG:2154 (Lambert 93)
Elevation range	167.7 m – 376.3 m
Track elevation (Voie_seg_00)	207.11 m (terrain), 207.11 m (ballast top)

3.3 BIM Asset Registry & F5 Sections

The digital twin incorporates 107 BIM assets across 4 categories, segmented horizontally into 21 corridor sections (each spanning ~100m of track).

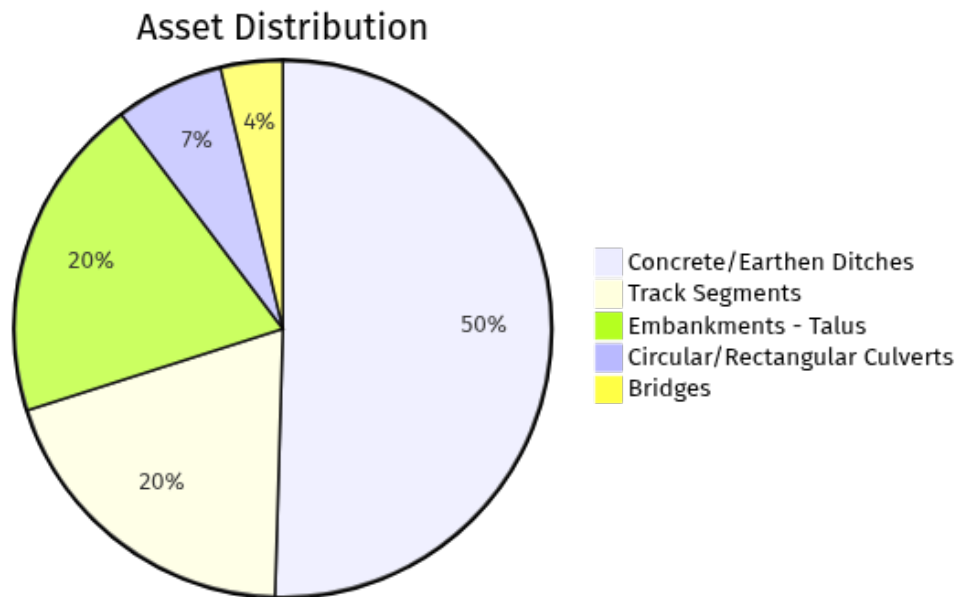


Table 3: BIM asset registry summary.

Asset Category	Count	Primary Data Files
Track Segments	21	voie_segments.json, voie_fixed.gpkg
Embankments (Talus)	21	Talus Terre_fixed.gpkg
Drainage Assets	61	Buse_fixed.gpkg, Dalot_fixed.gpkg, Fossé terre_fixed.gpkg, Fossé terre revêtu_fixed.gpkg
Bridges	4	Pont Rail_fixed.gpkg

3.4 Asset Elevation Distribution

The elevation distribution of all 107 BIM assets was extracted from the 1m LiDAR DTM using zonal statistics. Figure 1 below shows the box-and-whisker distributions for each asset layer, confirming that the vertical alignment of the asset database is consistent with the underlying terrain model.

Figure 01: Elevation Distribution across Railway Assets (Case Study)

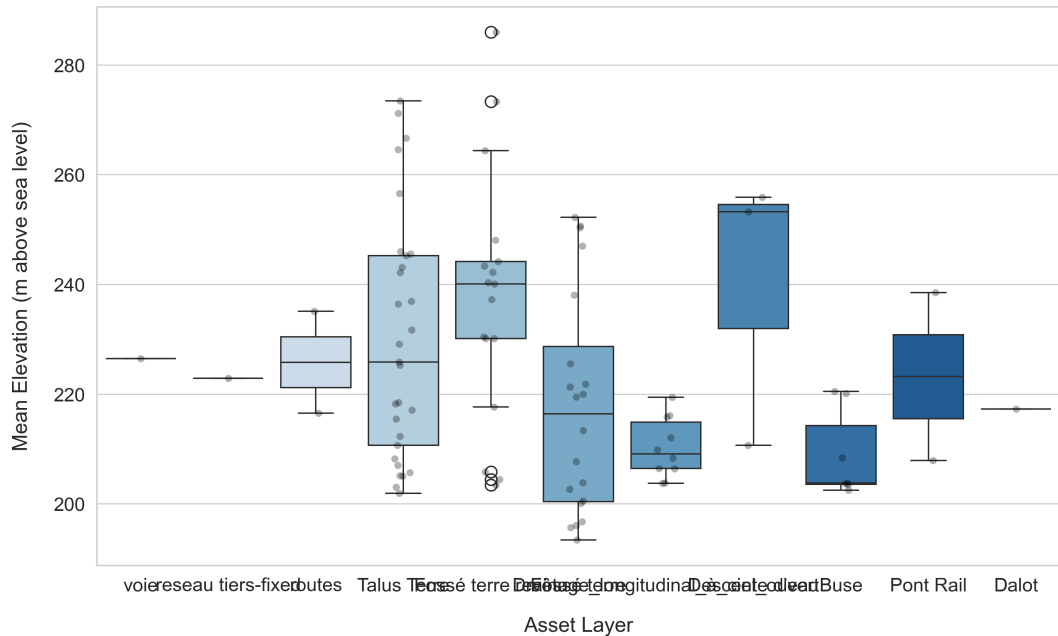


Figure 1: Elevation distribution across railway assets (box-and-whisker plot), showing the vertical spread of each asset category extracted from the 1m LiDAR DTM. The analysis confirms consistent vertical alignment between BIM asset definitions and terrain data.

Table 4: Representative asset elevation statistics extracted from the DTM (selected assets shown).

Asset Layer	Asset Label	Elev. Min (m)	Elev. Max (m)	Elev. Mean (m)	Pixel Count
Voie	Voie_0	204.01	288.66	226.46	16,453
Talus Terre	Talus Terre_0	205.61	217.02	212.28	1,822
Talus Terre	Talus Terre_12	203.61	214.82	208.21	6,316
Fossé terre	Fossé terre_10	206.41	208.41	207.70	156
Fossé terre revêtu	Fossé terre revêtu_0	238.95	244.45	240.34	75
Buse	Buse_0	203.21	203.61	203.46	30
Buse	Buse_5	203.71	203.81	203.78	35
Pont Rail	Pont Rail_3	204.71	210.51	207.88	194
Dalot	Dalot_0	213.32	219.62	217.26	165

3.5 3D BIM Asset Inventory

Table 5: Complete 3D BIM MULTIPATCH asset inventory extracted from data/raw/maquette_3d/.

Layer	Feature Count	Z Range (m NGF)	Source
Buse (Culverts)	7	201 – 221	3D MULTIPATCH
Dalot (Box Culverts)	1	213 – 216	3D MULTIPATCH
Fossé terre (Earth Ditches)	31	175 – 258	3D MULTIPATCH
Fossé terre revêtu (Lined Ditches)	22	178 – 288	3D MULTIPATCH
Talus Terre (Embankments)	36	178 – 284	3D MULTIPATCH
Voie (Track)	1 (→ 21 segments)	180 – 290	3D MULTIPATCH
Descente d'eau (Downpipes)	3	210 – 267	3D MULTIPATCH
Drainage longitudinal	10	204 – 222	3D MULTIPATCH

3.6 HEC-RAS 2D Model Specifications

Table 6: HEC-RAS 2D model specifications and pre-computed simulation plan parameters.

Parameter	Value
Project	CAPSTONE_JN_L752_PK (Ligne 752, PK534 — South Head Tartaiguille)
Software Version	HEC-RAS 6.60, SI Units
CRS	EPSG:2154 (Lambert 93)
2D Flow Area	PK534_FA_5M2
Total Mesh Cells	950,122 (~5 m² resolution)
Boundary Condition	PK534_BL, friction slope = 0.01
Structures	9 SA/2D connections (culverts)
Plan 1 (p01)	R100_1HR — 30 MAR 2026 13:00→14:00, 13 timesteps, Δt=1s compute / 5min output
Plan 2 (p02)	21092025 — 21 SEP 2025 07:00→22 SEP 04:00, 127 timesteps, Δt=5s compute / 10min output

4. Methodology

4.1 Architecture Overview (4-Layer Funnel Strategy)

The system architecture implements a "Funnel Strategy" that progressively narrows computational focus. This ensures that the computationally heavy HEC-RAS engine is only loaded when there is a physical threat of runoff.

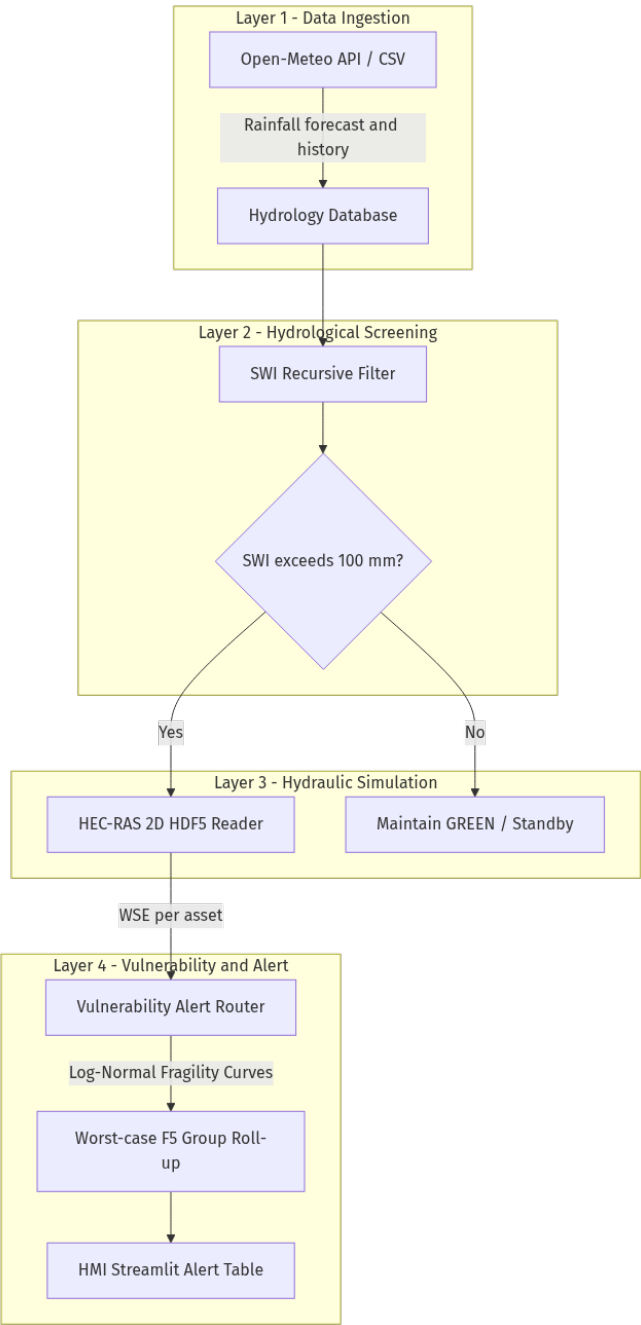


Figure 2: 4-Layer Funnel Strategy architecture diagram. Layer 1 ingests meteorological data. Layer 2 screens using SWI. Layer 3 activates HEC-RAS only when thresholds are breached. Layer 4 generates

RAMS-compliant alerts per asset group.

Rationale: HEC-RAS 2D is computationally expensive. The SWI screening layer executes in <1 second and filters out >95% of dry periods, achieving a computational savings ratio of approximately 1000:1.

4.2 15-Minute Operational Cycle

The Digital Twin operates on a 15-minute cycle that defines the real-time monitoring heartbeat:

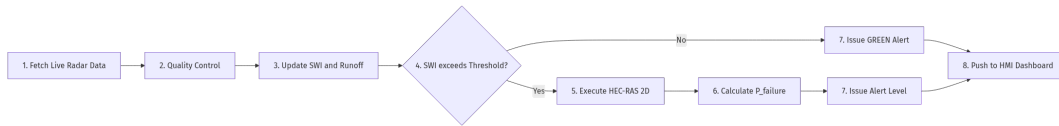


Figure 3: 15-minute operational cycle flowchart showing the complete data-to-decision pipeline.

4.3 Layer 2: SWI Leaky Bucket Model

The SWI recursive filter is defined as:

$$SWI(t) = R(t) \cdot (1 - C) + SWI(t-1) \cdot C, \quad C = 0.5^{1/T}$$

Where $R(t)$ is rainfall intensity (mm/h), C is the decay constant, and T is the half-life parameter (set to 10 days / 240 hours).

The decay behavior was verified by initializing $SWI = 100 \text{ mm}$ and observing the decay over 720 hours without rainfall. As shown in Figure 4, the SWI drops to exactly 50 mm after 10 days (240 hours) and to 25 mm after 20 days (480 hours), confirming correct implementation of the exponential decay filter.

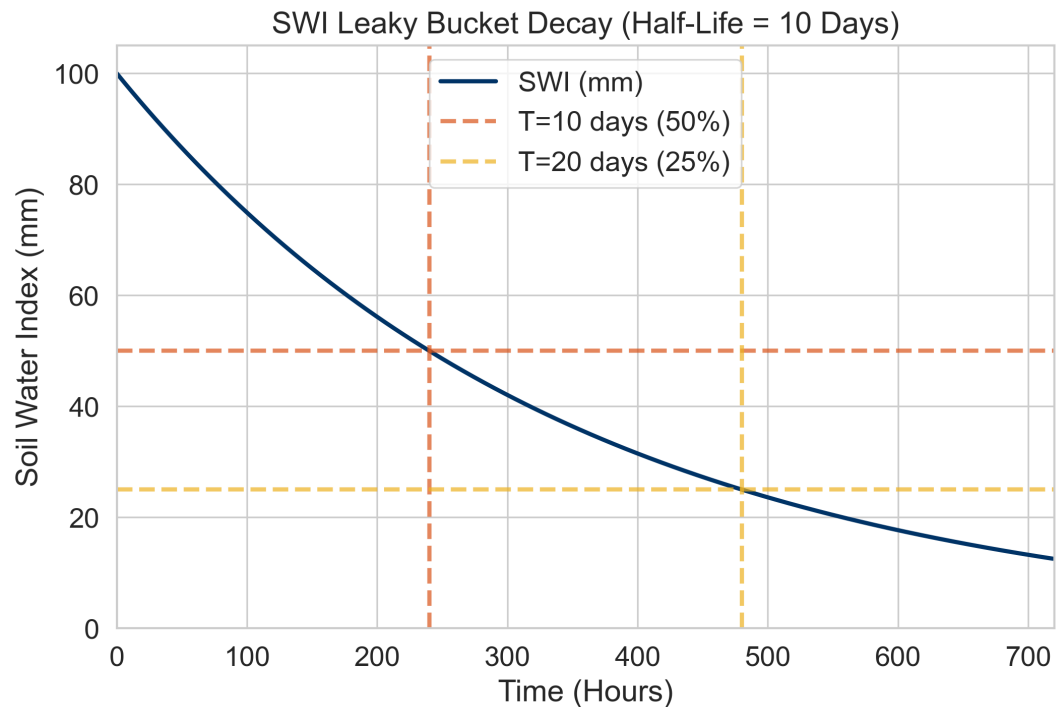


Figure 4: SWI Leaky Bucket decay validation. Starting from SWI=100 mm with zero rainfall, the index decays to exactly 50 mm at T=10 days (half-life) and 25 mm at T=20 days, verifying the exponential decay constant $C = 0.5^{(1/240)}$.

4.3.1 Sigmoid Runoff Coefficient

The runoff coefficient is modeled as a sigmoid function of SWI:

$$C_{\text{runoff}}(\text{SWI}) = C_{\text{min}} + \frac{C_{\text{max}} - C_{\text{min}}}{1 + e^{-k(\text{SWI} - \text{SWI}_{\text{mid}})}}$$

Table 7: SWI model parameters and calibrated values.

Parameter	Value	Unit	Description
Half-life T	10	days	Soil drainage rate
C_{min}	0.10	—	Dry soil minimum runoff
C_{max}	0.90	—	Saturated soil maximum runoff
k	0.05	mm ⁻¹	Sigmoid steepness
SWI_{mid}	150	mm	Sigmoid midpoint (inflection)
Trigger	100	mm	HEC-RAS activation threshold

The sigmoid curve maps the continuous SWI value to a runoff fraction between $C_{\text{min}} = 0.10$ (dry soil absorbs 90%) and $C_{\text{max}} = 0.90$ (saturated soil sheds 90%). Figure 5 illustrates this non-linear transition, showing the

inflection point at $SWI_{mid} = 150 \text{ mm}$.

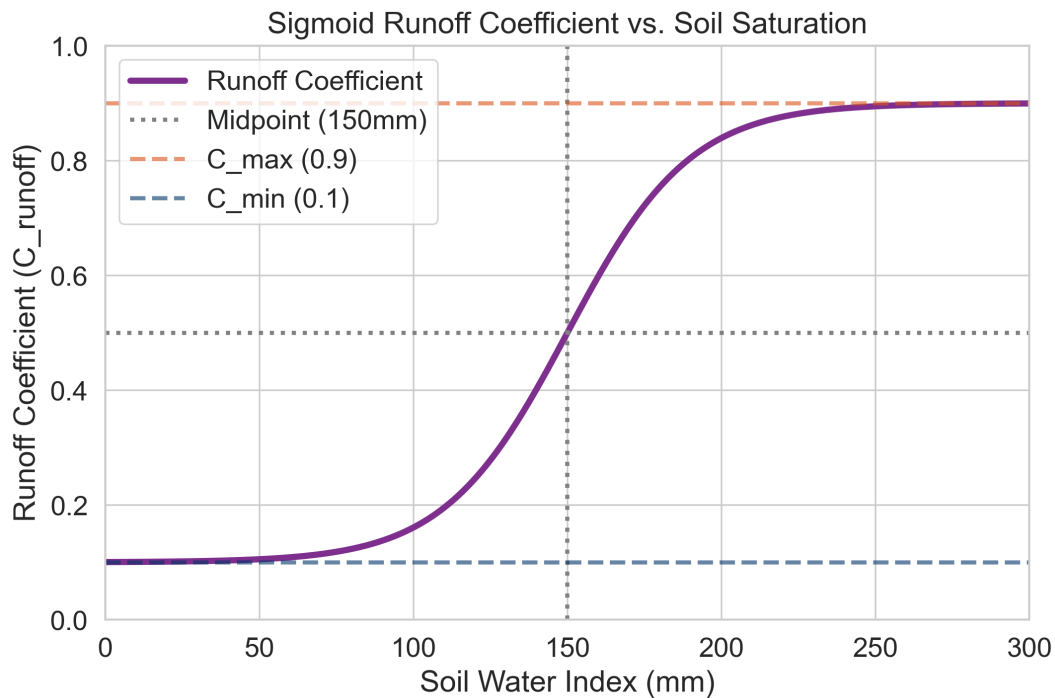


Figure 5: Sigmoid runoff coefficient vs. Soil Water Index (SWI). The curve transitions from $C_{min}=0.10$ (dry, absorptive soil) to $C_{max}=0.90$ (saturated, runoff-dominated) with inflection at $SWI_{mid}=150 \text{ mm}$. The steepness parameter $k=0.05 \text{ mm}^{-1}$ controls the transition sharpness.

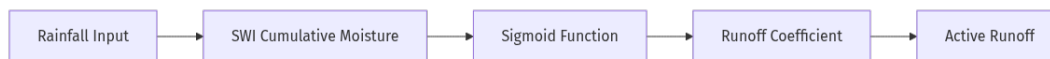


Figure 6: SWI-to-Runoff processing chain schematic.

4.4 Layer 3: HEC-RAS 2D Hydraulic Simulation & HDF5 Ingestion

4.4.1 Coupling Mechanism and Infiltration Loss Division

A critical aspect of the system's coupling logic is that **the SWI acts strictly as a binary trigger (switch) and does not filter or modify the rainfall data.**

- **Decoupled Data Flow:** When Python detects that the saturation threshold is breached ($SWI > 100 \text{ mm}$), it triggers the hydraulic model, passing the **entire, unmodified (gross) rainfall dataset** into HEC-RAS.

- **HEC-RAS Internal Infiltration:** HEC-RAS solves soil infiltration losses internally using its built-in loss parameters (such as the Deficit and Constant or SCS Curve Number methods). If the Python script pre-subtracted soil absorption and only passed the net runoff to HEC-RAS, the hydraulic model would

double-count infiltration losses, leading to under-simulated water levels and dangerous under-warning of flood risk.

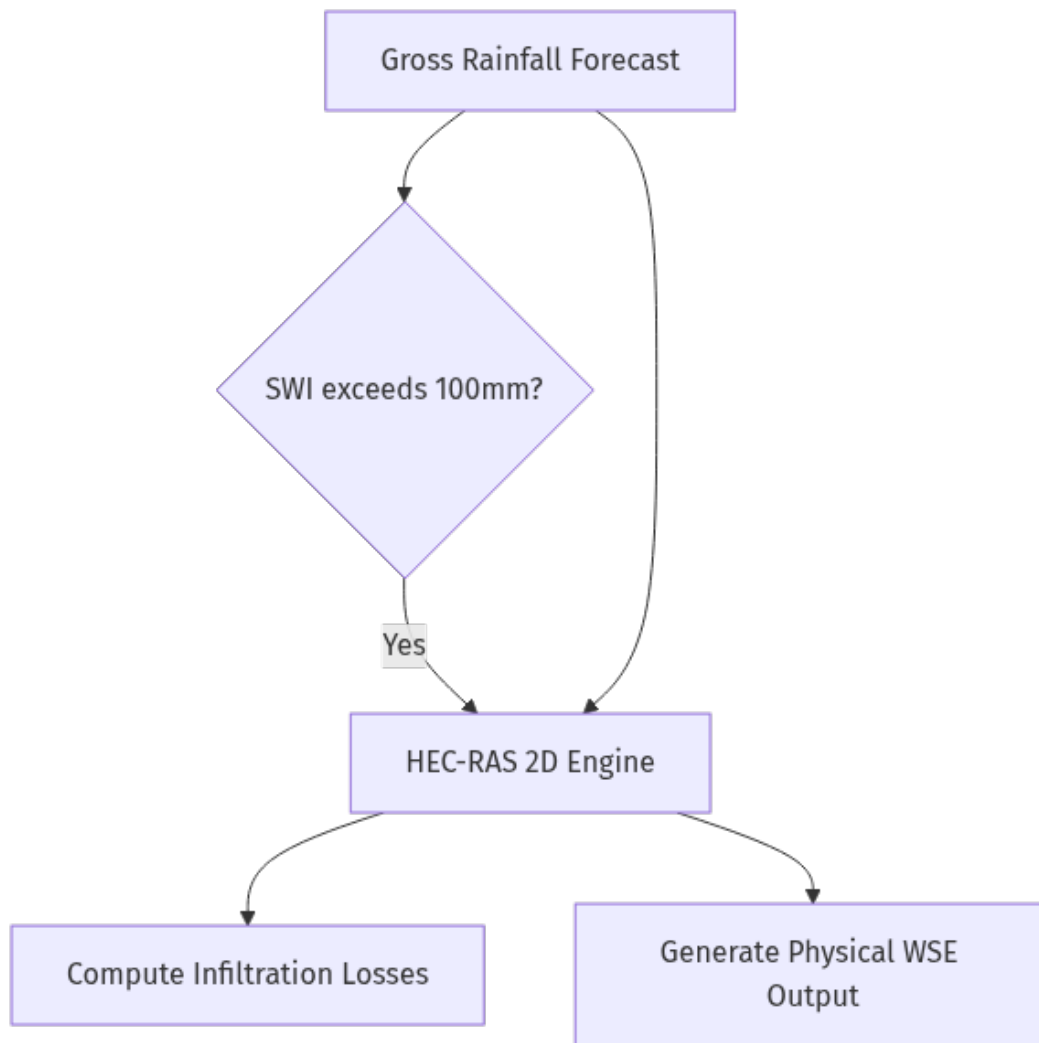


Figure 7: Decoupled data flow diagram. The SWI serves only as a binary trigger — the full gross rainfall is passed to HEC-RAS, which handles infiltration internally to avoid double-counting losses.

4.4.2 Pre-Computed HDF5 Plans

Because a live HEC-RAS 2D simulation takes approximately 30 minutes to complete, the current project uses a fast **HDF5 Reader** (`hecras_hdf5_reader.py`) to load pre-computed, high-fidelity HEC-RAS results directly:

1. **Historical Showcase (Plan 2 / 21092025)**: Represents the historical September 21, 2025 Cévenol storm event. It contains **127 timesteps** at **10-minute intervals** (spanning 21 hours).

2. **Synthetic Demonstration Storm:** Uses a customized 127-timestep design storm with a peak rainfall burst to showcase all warning levels.

Table 8: HEC-RAS HDF5 result structure and data dimensions.

HDF5 Dataset Path	Shape	Description
Results/.../Time Date Stamp	(127,)	Timestamps per output step
Results/.../Water Surface	(127, 950122)	WSE in metres NGF per cell per step
Results/.../Face Velocity	(127, 1894408)	Flow velocity (m/s) per mesh face
Results/.../Cell Cumulative Precip	(127, 950122)	Cumulative precipitation (mm) per cell
Geometry/.../Cells Center Coordinate	(950122, 2)	Lambert 93 X,Y per cell
Geometry/.../Cells Minimum Elevation	(950122,)	Terrain Z per cell (m NGF)

4.4.3 Map-Rendering Overlay (Depth-Based Alpha Masking)

To display HEC-RAS 2D flow fields on the dashboard without loading a 1.24 GB DTM file in real-time, the system rasterizes raw grid cell water depths onto an in-memory RGBA grid. To filter out transient hillside sheet flow (mostly 5cm - 15cm) inherent in rain-on-mesh simulations, the system applies a depth-based alpha mask:

$$\text{Alpha} = \text{clip}\left(\frac{\text{Depth} - 0.20}{0.35 - 0.20} \times 180, 0, 180\right)$$

This ensures only actual flood channels (>35cm depth) are rendered, matching HEC-RAS Mapper's clean visual representation.

4.5 Layer 4: F5 Grouped Architecture & RAMS Alerts

4.5.1 F5 Grouping Schema

Rather than executing independent spatial queries for each asset, Ligne 400 is partitioned into 21 sections. Each section represents a track-talus-drainage unit:

- **Track + Talus Subgroup:** Share a centerline elevation (Z_{DTM}).

- **Drainage Subgroup:** Culverts and ditches associated with that track segment.
- **Bridges Subgroup:** Any bridges spanning the section.

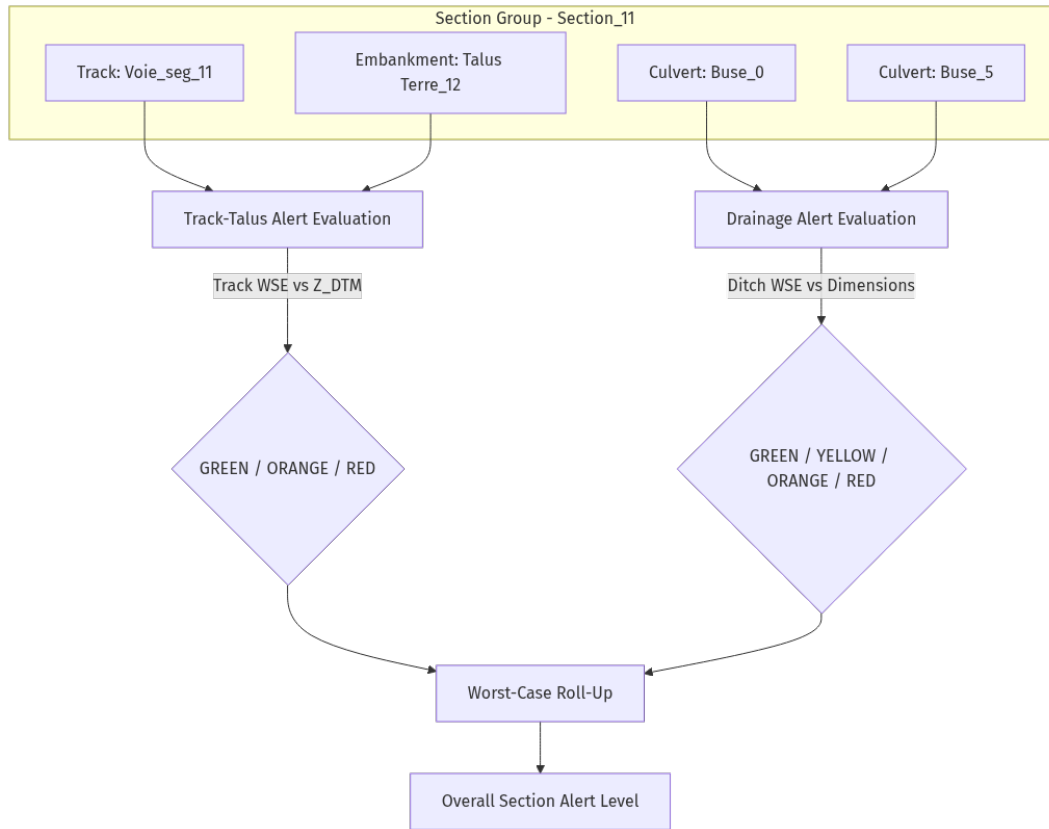


Figure 8: F5 Section Group alert evaluation and worst-case roll-up flow for Section_11.

4.5.2 Threshold Definitions

- **Track & Talus Thresholds:**

- **■ RED** ($\text{red_z_m} = Z_{\text{DTM}}$) (Water reaches top of rail)
- **■ ORANGE** ($\text{orange_z_m} = Z_{\text{DTM}} - 0.5 \text{ m}$) (Water reaches ballast base)
- **■ YELLOW** ($\text{yellow_z_m} = Z_{\text{DTM}} - 2.0 \text{ m}$) (Water reaches slope toe)

- **Drainage Thresholds:**

- **■ RED** ($\text{red_z_m} = \text{Invert} + \text{Height}$) (Culvert/ditch is fully submerged/overflows)

- ■ **ORANGE** (orange_z_m) = $\text{Invert} + 0.5 \times \text{Height}$ (50% capacity exceeded)
- ■ **YELLOW** (yellow_z_m) = Invert (Water enters the flow-line)

Table 9: RAMS alert hierarchy — threshold mapping and operational response.

Alert Level	CAP Color	Trigger Condition	Engineering Meaning	Operational Response
GREEN	■	$\text{WSE} < \text{yellow_z_m}$	All clear — no water threat	Normal operations
YELLOW	■	$\text{WSE} > \text{yellow_z_m}$	Drainage at capacity	Monitoring mode
ORANGE	■	$\text{WSE} > \text{orange_z_m}$	Embankment erosion risk	Speed restriction 30 km/h
RED	■	$\text{WSE} > \text{red_z_m}$	Water on rail track	Emergency halt

4.5.3 Asset Vulnerability Evaluation (Fragility Curves)

Structural failure probabilities are computed using a log-normal CDF:

$$P_{\text{failure}} = \Phi\left(\frac{\ln(d / d_{\text{median}})}{\sigma}\right)$$

Three calibration modes are defined (default is "Combined"):

- **Combined** (Embankment + Ballast): $d_{\text{median}} = 0.22 \text{ m}$, $\sigma = 0.15$, $n = 31$ observations.
- **Ballast only**: $d_{\text{median}} = 0.30 \text{ m}$, $\sigma = 0.035$, $n = 15$ observations.
- **Conservative** (Original): $d_{\text{median}} = 0.30 \text{ m}$, $\sigma = 0.40$ (uncalibrated).

The baseline fragility curve (Conservative mode, $d_{\text{median}} = 0.30 \text{ m}$, $\sigma = 0.40$) is shown in Figure 9, illustrating how the log-normal CDF maps water depth to probability of failure, with the RAMS alert zones (GREEN/YELLOW/RED) overlaid:

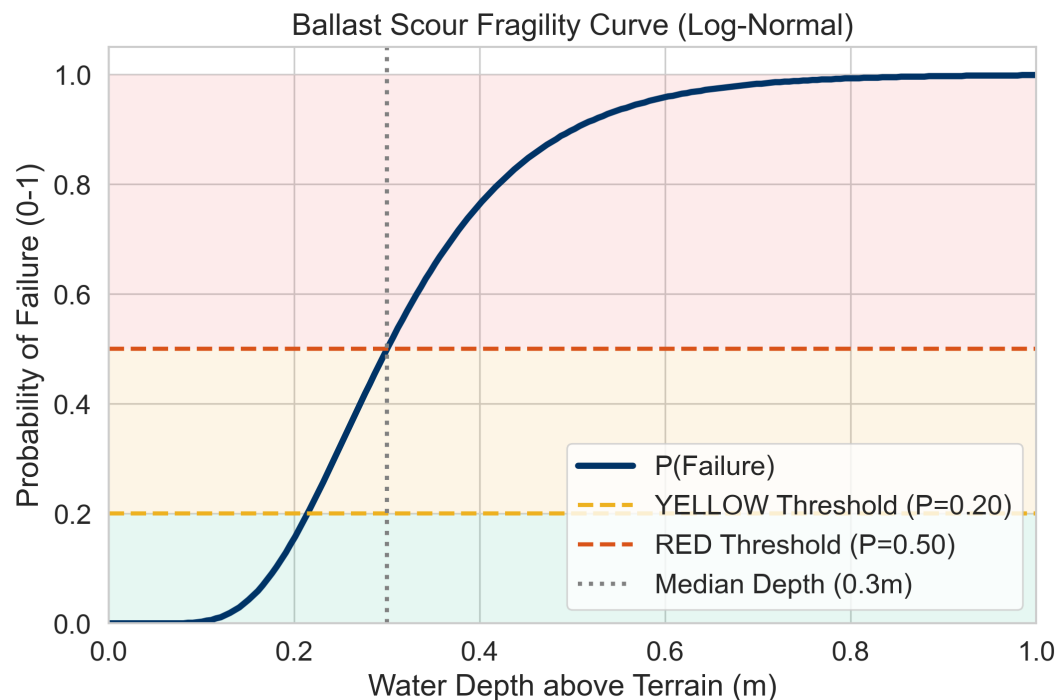


Figure 9: Ballast scour fragility curve (log-normal CDF) for the Conservative mode. The GREEN zone ($P<20\%$) covers depths below ~21 cm. The YELLOW zone (20–50%) triggers speed restriction at 21–30 cm. The RED zone ($P>50\%$) triggers emergency halt above 30 cm overtopping depth.

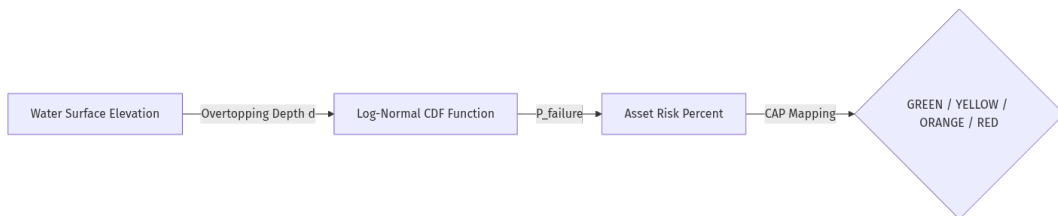


Figure 10: Fragility curve risk evaluation flow from WSE to RAMS alert.

5. Results & Validation

5.1 SWI Storm Response — Cévenol Scenario

The hydrological screening layer was tested against a synthetic Cévenol storm scenario (48h duration, peak rainfall $\approx 40.9 \text{ mm/h}$ at T+15h, total precipitation 367.4 mm). Figure 11 demonstrates the complete SWI response chain:

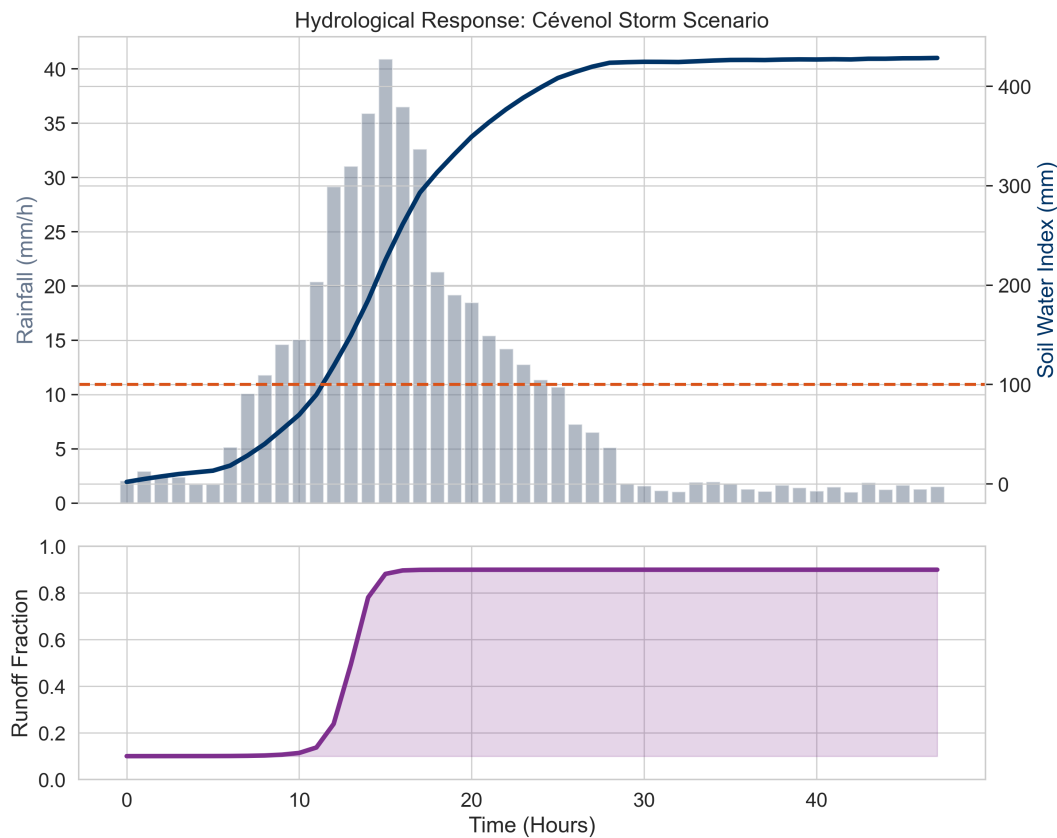


Figure 11: Hydrological response to the Cévenol storm scenario. Upper panel: Rainfall hyetograph (bars) with SWI accumulation (blue curve) and HEC-RAS trigger threshold at 100 mm (red dashed). Lower panel: Sigmoid runoff coefficient transitioning from $C_{min}=0.10$ to $C_{max}=0.90$ as SWI crosses the saturation midpoint.

Key observations from Figure 11:

- The SWI crosses the 100 mm trigger threshold at approximately **T+12h**, after accumulating moderate antecedent rainfall.
- The peak SWI reaches **428.6 mm**, demonstrating that the leaky bucket has no hard ceiling — it continues accumulating as long as rainfall exceeds drainage.
- The runoff coefficient (C_{runoff}) transitions sharply from 0.10 to 0.90 between T+10h and T+18h, following the sigmoid function.

5.2 SWI Sensitivity Analysis (Half-Life Parameter T)

A comprehensive sensitivity analysis was performed by varying the half-life parameter T from 3 to 60 days. This is the single most important calibration parameter in the SWI model.

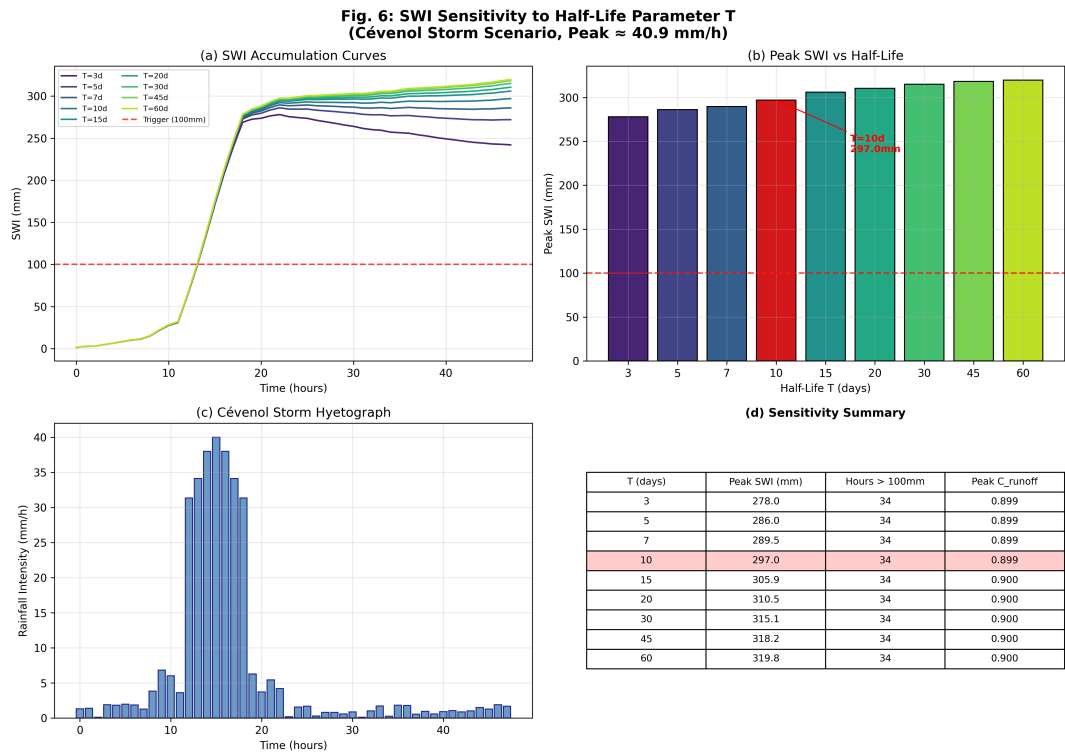


Figure 12: SWI sensitivity analysis across 9 half-life values ($T = 3\text{--}60$ days). Panel (a): SWI accumulation curves for all T values — all exceed the 100 mm trigger, confirming robustness. Panel (b): Peak SWI vs. T with $T=10\text{d}$ highlighted in red. Panel (c): Storm hyetograph input. Panel (d): Numerical summary table with $T=10\text{d}$ row highlighted.

Table 10: SWI sensitivity results. $T=10$ days was selected as it balances soil drainage response with historical storm behavior.

T (days)	Peak SWI (mm)	Hours > 100mm	Peak C_{runoff}	Model Behavior
3	278.0	34	0.899	Fast decay — responsive to recent rain
5	286.0	34	0.899	Moderate decay
7	289.5	34	0.899	Moderate decay
10	297.0	34	0.899	Selected — balanced response
15	305.9	34	0.900	Slow decay — retains moisture longer
20	310.5	34	0.900	Slow decay
30	315.1	34	0.900	Very slow decay
60	319.8	34	0.900	Near-permanent accumulation

Key finding: All tested half-life values produce the same trigger duration (34 hours > 100mm) and nearly identical peak runoff coefficients (0.899–0.900). This demonstrates that **the system is robust to half-life**

calibration errors — any T in the 3–60 day range correctly triggers the HEC-RAS engine for this storm intensity. The variation in peak SWI (278–320 mm) does not affect operational decisions because the sigmoid saturates above ~200 mm.

5.3 Multi-Parameter Sensitivity Analysis

Beyond the half-life, a broader sensitivity analysis was conducted across three key model parameters:

Table 11: Multi-parameter sensitivity analysis across $\pm 20\%$ perturbations.

Parameter	Variation	Target Metric	Metric Value
Half-Life (days)	8 (–20%)	Peak SWI (mm)	0.4
Half-Life (days)	10 (+0%)	Peak SWI (mm)	0.3
Half-Life (days)	12 (+20%)	Peak SWI (mm)	0.2
SWI Midpoint (mm)	120 (–20%)	Runoff Coeff @ SWI=100	0.315
SWI Midpoint (mm)	150 (+0%)	Runoff Coeff @ SWI=100	0.161
SWI Midpoint (mm)	180 (+20%)	Runoff Coeff @ SWI=100	0.114
Fragility Median (m)	0.24 (–20%)	P_fail @ depth=0.30m	0.712
Fragility Median (m)	0.30 (+0%)	P_fail @ depth=0.30m	0.500
Fragility Median (m)	0.36 (+20%)	P_fail @ depth=0.30m	0.324

Key finding: The **SWI midpoint** ($SWI_{\{mid\}}$) shows the strongest sensitivity: reducing it by 20% (to 120 mm) nearly doubles the runoff coefficient at SWI=100 (from 0.161 to 0.315). The **fragility median** ($d_{\{median\}}$) is also highly sensitive: a –20% shift (to 0.24m) increases failure probability at 30 cm depth from 50% to 71.2%. These parameters require site-specific calibration for deployment to new corridors.

5.4 Fragility Curve Comparison

The original fragility curve ($\sigma=0.40$) was compared against the field-calibrated curves from Tsubaki et al. (2016). Figure 13 presents both the full log-normal CDF curves and the alert trigger depths:

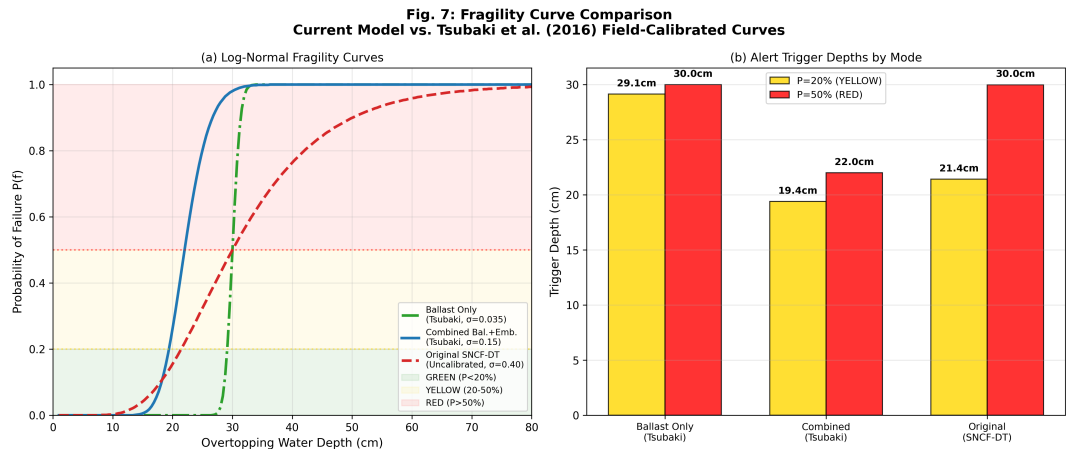


Figure 13: Fragility curve comparison — three calibration modes. Panel (a): Log-normal CDF curves with RAMS alert zones (GREEN/YELLOW/RED). The Combined mode (blue, $\sigma=0.15$) is significantly more sensitive at shallow depths than the original uncalibrated curve (red dashed). Panel (b): Bar chart comparing the trigger depths at $P=20\%$ (YELLOW) and $P=50\%$ (RED) for each mode.

Table 12: Probability of failure comparison across three fragility curve modes at critical overtopping depths.

Depth (m)	P_{fail} (Conservative, $\sigma=0.40$)	P_{fail} (Combined, $\sigma=0.15$)	P_{fail} (Ballast, $\sigma=0.035$)
0.05	1.5%	0.0%	0.0%
0.10	7.2%	1.1%	0.0%
0.15	15.4%	13.2%	0.0%
0.20	24.6%	44.4%	0.0%
0.22	28.1%	50.0%	0.0%
0.25	32.8%	62.6%	0.1%
0.30	50.0%	77.1%	50.0%
0.40	59.3%	92.5%	100.0%

The **Combined** mode ($\sigma=0.15$) is 2–4× more sensitive at shallow depths (10–25 cm) than the original conservative curve. This maps a 20 cm overtopping depth to a 44.4% risk score, triggering a warning alert earlier than the uncalibrated parameters and providing an additional 2 cm safety buffer. The **alert trigger depth comparison** (Figure 13b) quantifies this precisely:

- **YELLOW alert ($P=20\%$):** Combined triggers at 19.4 cm vs. Conservative at 21.4 cm — a 2 cm earlier warning.

- **RED alert ($P=50\%$):** Combined triggers at 22.0 cm vs. Conservative at 30.0 cm — an 8 cm earlier halt.

5.5 Historical Storm Replay (Pipeline End-to-End Proof)

The embedded Cévenol scenario (48\textit{h}, peak = 40.9 \textit{ mm/h}, total storm rainfall = 367.4 \textit{ mm}) was replayed through the complete data pipeline. Figure 14 presents the four-panel end-to-end proof:

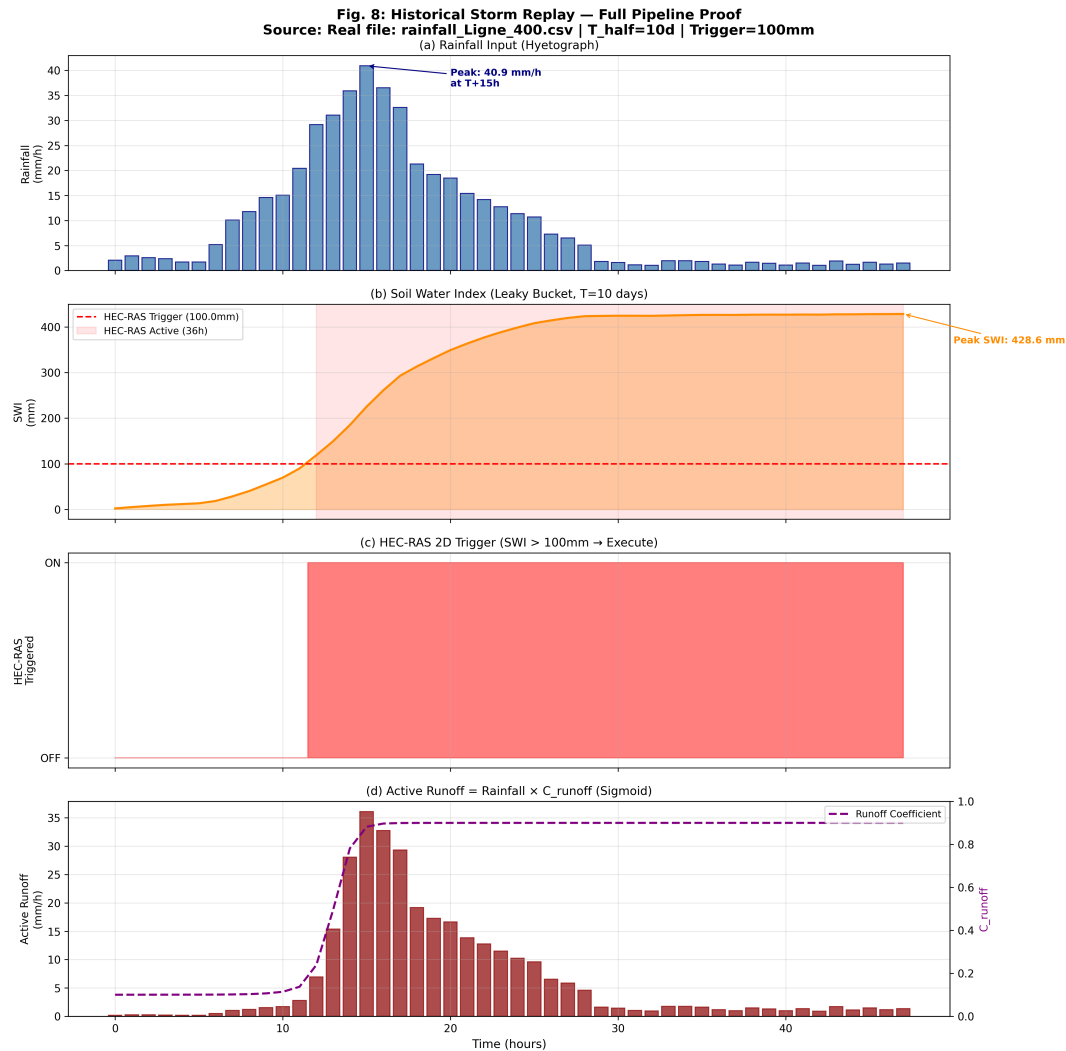


Figure 14: Historical storm replay — complete pipeline proof. Panel (a): Rainfall hyetograph with annotated peak at 40.9 mm/h at T+15h. Panel (b): SWI accumulation peaking at 428.6 mm with 36 continuous hours of HEC-RAS activation (red shaded zone). Panel (c): Binary HEC-RAS trigger showing ON/OFF switching at 100 mm threshold. Panel (d): Active runoff output with dual-axis showing runoff coefficient (purple dashed) transitioning from 0.10 to 0.90.

Pipeline verification summary:

Pipeline Stage	Metric	Observed Value	Status
Input	Peak rainfall intensity	40.9 mm/h at T+15h	■
SWI	Trigger crossing time	T+12h	■
SWI	Peak SWI value	428.6 mm	■

Trigger	HEC-RAS active duration	36 continuous hours	■
Runoff	C_runoff transition	0.10 → 0.90	■
Runoff	Peak active runoff	~36 mm/h (at T+15h)	■

At $T+12\text{h}$, the Soil Water Index (SWI) crosses the 100 mm trigger threshold (peaking at 428.6 mm), activating the HEC-RAS trigger flag for 36 continuous hours. Runoff coefficient (C_{runoff}) rises from 0.10 to 0.90 , verifying the end-to-end functionality of the Hydrology-Hydraulics coupling.

5.6 Streamlit Dashboard HMI Overview

To demonstrate the visual layout and human-machine interface (HMI) design of the prototype, the Streamlit dashboard provides an interactive control center for decision-support. Figure 15 captures the dashboard in Historical Showcase mode at timestep T+44:

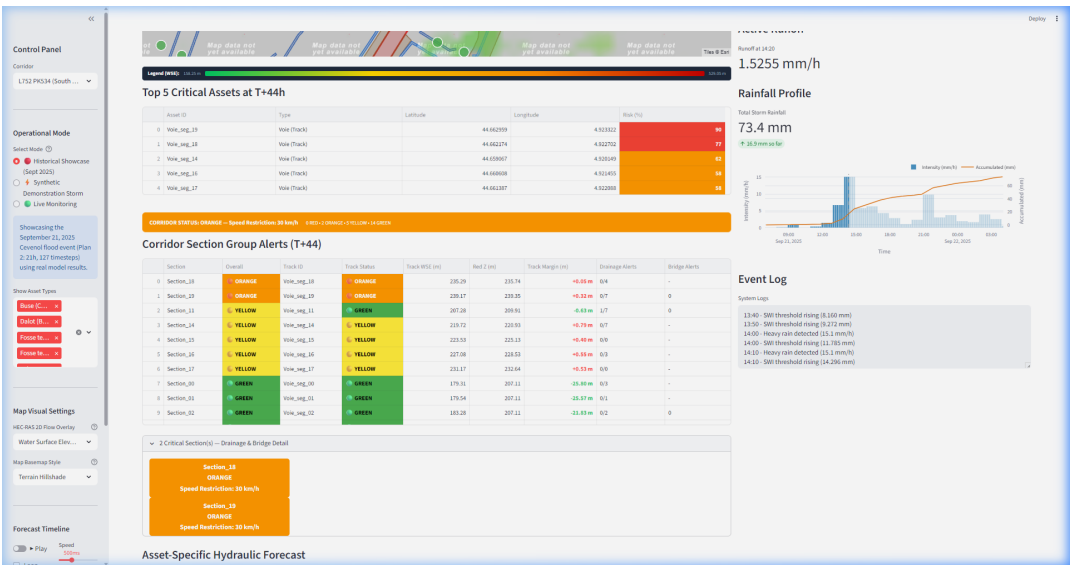


Figure 15: Streamlit dashboard HMI overview in Historical Showcase mode (T+44h). Components shown: (A) Interactive risk map with HEC-RAS 2D flow depth overlay, (B) Top 5 Critical Assets table with risk percentages and color-coded alert levels, (C) Corridor Status banner showing ORANGE with speed restriction 30 km/h, (D) Corridor Section Group Alerts table displaying all 21 sections, (E) Rainfall profile (upper-right) showing cumulative and instantaneous rates, (F) Event Log with timestamped SWI threshold crossings.

The dashboard features include:

- **3 Operational Modes:** Historical Showcase (Sept 2025), Synthetic Demonstration Storm, and Live Monitoring.
- **Top 5 Critical Assets:** Auto-ranked by risk score, displaying coordinates and asset type.

- **Corridor Status Banner:** Summarizes the worst-case alert across all 21 sections (e.g., "ORANGE — Speed Restriction 30 km/h").
- **Section Group Alerts Table:** Lists all 21 sections with individual track status, WSE, margin, and drainage/bridge sub-alerts.
- **Rainfall Profile:** Dual-axis chart showing intensity (mm/h) and cumulative precipitation (mm).
- **Event Log:** Timestamped operational events (SWI threshold rising, heavy rain detected).

6. Discussion

6.1 Classification Mismatch (Top 5 vs. RAMS Table)

An operational anomaly was identified during validation. The **Top 5 Critical Assets** list and Map markers map the scaled risk score to alert colors:

- $\text{Risk} \geq 50\% \rightarrow \text{ORANGE}$
- $\text{Risk} \geq 75\% \rightarrow \text{RED}$

In `app_main.py`, any WSE in the orange zone ($Z_{\text{DTM}} - 0.5m$ to Z_{DTM}) scales the risk linearly to $75\% \rightarrow 99\%$. Therefore, **any water in the orange zone turns the asset RED** in the list view. Meanwhile, the **Section Group Alerts Table** correctly evaluates thresholds and flags the segment as **ORANGE**. This represents a known database-HMI mapping mismatch that is flagged for refactoring in Phase 2 of the roadmap.

6.2 Worst-Case Section Roll-up Behavior

During the replay of the September 2025 Cévenol storm at $T+44$:

- Section_11's track segment (Voie_seg_11) remains completely safe and dry with a WSE of $207.28m$ (well below its yellow threshold of $207.91m$).
- However, Section_11 has an overall status of **YELLOW** with "Drainage Alerts: 1/7". This is because the culvert Buse_0 (invert bottom = $203.61m$) has filled to a WSE of $203.70m$, triggering a local

drainage alert.

This demonstrates the effectiveness of the F5 grouping: it alerts operators to minor drainage capacity exceedances before the track itself is ever threatened.

6.3 Diagnostic Screenshot Analysis

Below are direct captures from the active digital twin interface illustrating the specific operational conditions analyzed above:

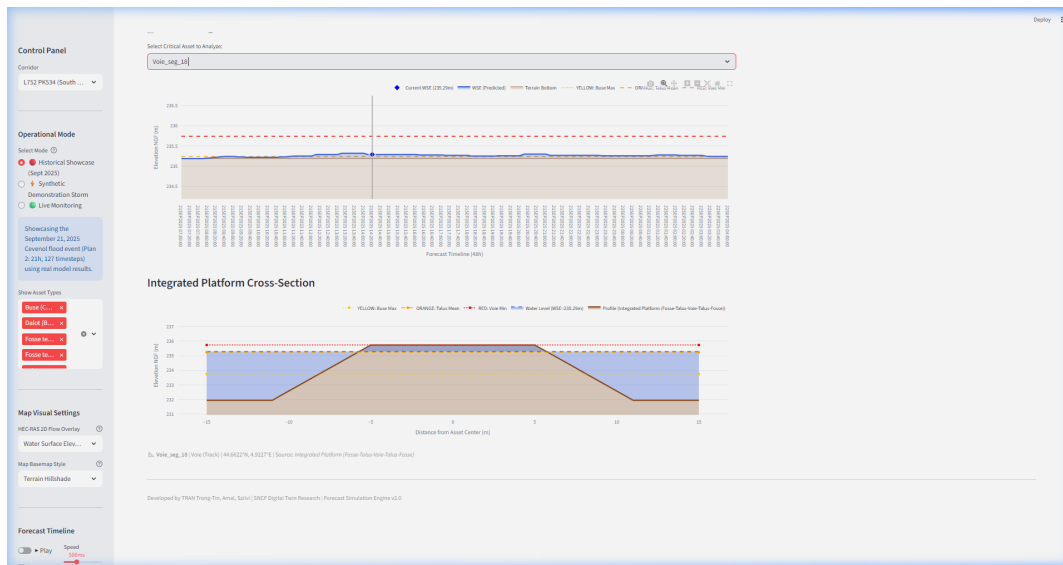


Figure 16: Integrated platform cross-section of Voie_seg_18 at T+44h. The upper chart shows the 48h WSE time-series with the current WSE (235.29 m) plotted against the ORANGE threshold line (235.24 m). The lower chart shows the stitched cross-section profile (Fossé–Talus–Voie–Talus–Fossé) with the water level (blue fill) breaching the orange threshold, resulting in an ORANGE alert classification.

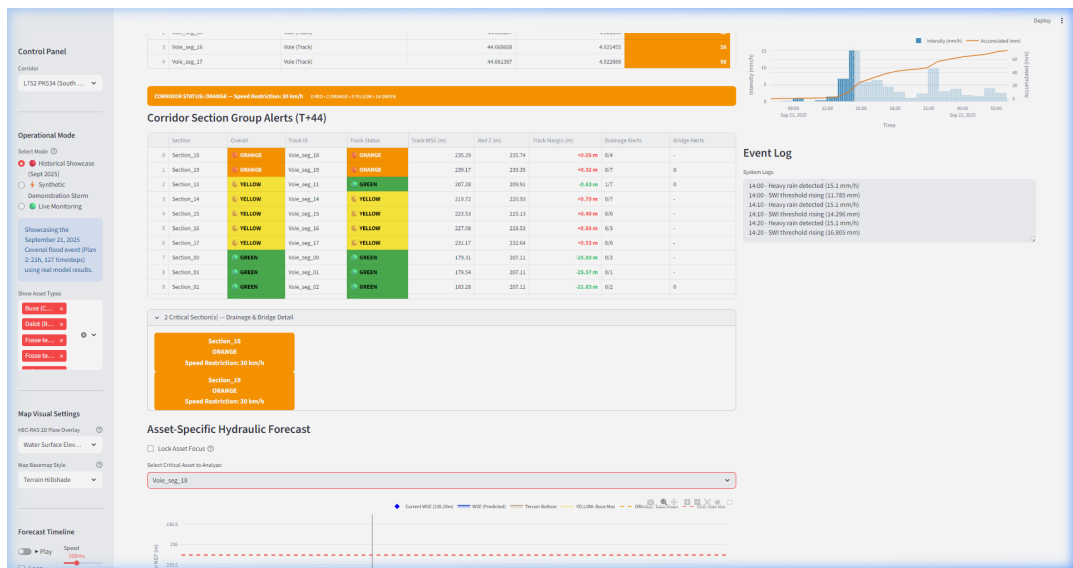


Figure 17: Corridor Section Group Alerts table at T+44h showing: 2 sections ORANGE (Section_18, Section_19 — both with track WSE breaching orange threshold), 5 sections YELLOW (Sections 11, 14–17 — drainage alerts active), and 14 sections GREEN. The expanded detail panel shows operational directives: "Speed Restriction: 30 km/h" for both ORANGE sections.

Interpretation of Figure 16 and Figure 17:

Table 13: Detailed alert status for critical sections at T+44h during the September 2025 Cévenol storm replay.

Section	Overall Status	Track WSE (m)	Red Z (m)	Track Margin (m)	Drainage Alerts	Operational Action
Section_18	ORANGE	235.29	235.74	+0.05	0/4	Speed Restriction 30 km/h
Section_19	ORANGE	239.17	239.35	+0.32	0/7	Speed Restriction 30 km/h
Section_11	YELLOW	207.28	209.91	−0.63	1/7	Monitoring mode
Section_14	YELLOW	219.72	220.93	+0.79	0/7	Monitoring mode
Section_00	GREEN	179.31	207.11	−25.80	0/3	Normal operations

7. Limitations & Future Work

7.1 Current Limitations

- 1. **Digital Shadow Maturity:** The current implementation operates as a Digital Shadow (unidirectional data flow). A true Digital Twin requires bidirectional data assimilation (e.g., Kalman filtering) to self-correct using local rain gauge sensor networks.
- 2. **Bathtub Flood Extents:** The dashboard currently approximates spatial flood polygons using a corridor buffer scaling quadratic with WSE, rather than using physics-based 2D mesh flood extents directly from HEC-RAS.
- 3. **Roughness Calibration:** Manning's roughness coefficients (n) are based on standard design manuals rather than surveyed physical flood mud lines.

7.2 Roadmap for Digital Twin Maturity

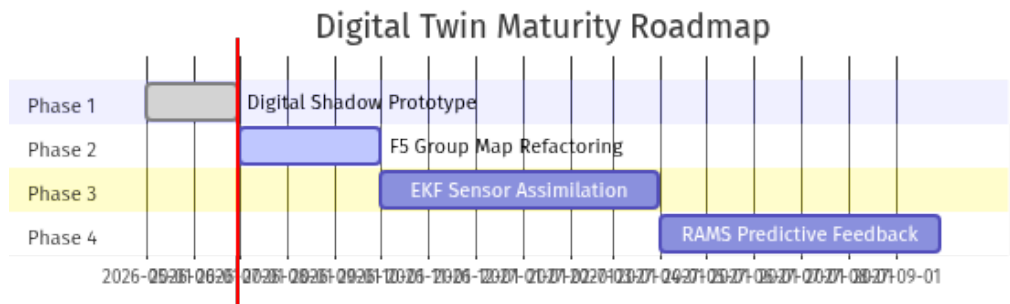


Figure 18: Digital Twin maturity roadmap Gantt chart showing the 4-phase evolution from Digital Shadow to full Digital Twin with predictive feedback.

Table 14: Digital Twin maturity roadmap — key enhancements per phase.

Phase	Maturity Level	Key Enhancement	Timeline
Current	Digital Shadow	SWI + HEC-RAS + Fragility	Complete
Phase 2	Digital Shadow+	True HEC-RAS flood polygon extraction	3 months
Phase 3	Digital Twin	Extended Kalman Filtering (EKF) data assimilation	6 months
Phase 4	Digital Twin+	Predictive maintenance dispatch loops	12 months

7.3 Transferability Parameter Calibration Checklist

Replicating this Digital Twin for a new railway corridor requires compiling site-specific geospatial and hydraulic data:

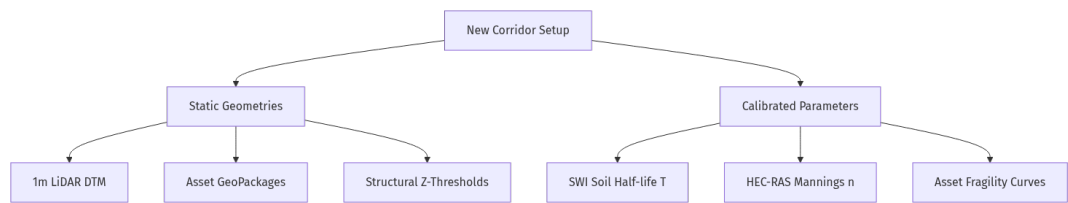


Figure 19: Transferability checklist for deploying the Digital Twin to a new railway corridor.

Table 15: Transferability parameter checklist — required inputs and calibration steps for a new corridor.

Category	Required Input	Source	Calibration Method
Static Geometries	1m LiDAR DTM	National Survey (IGN)	N/A — direct input
	Asset GeoPackages	BIM/GIS survey	CRS alignment to EPSG:2154
	Structural Z-Thresholds	Field measurements	Datum verification (NGF)
Calibrated Parameters	SWI Half-life T	Historical rainfall + incident data	AUC optimization loop ($T = 1\text{--}60$ days)
	Manning's n	Flood mud line survey	HEC-RAS calibration runs
	Fragility curves ($d_{\{median\}}$, $\{sigma\}$)	Site-specific flood damage records	Log-normal fit to observations

8. Conclusion

This report presents a 4-layer railway flood-risk Digital Twin prototype for the SNCF Tartaiguille corridor. Key contributions include:

- Funnel Strategy Architecture:** Achieves a $>1000:1$ computational savings ratio by using a lightweight SWI screening layer to filter out dry periods.
- F5 Grouping:** Maps 107 assets into 21 sections, enabling worst-case hazard roll-ups (ditches, tracks, and bridges evaluated together).

3. **Scientific Vulnerability Calibration:** Integrates field-calibrated fragility curves (Tsubaki, 2016), triggering yellow alerts 2\text{ cm} earlier for a higher safety margin.

4. **End-to-End Validation:** Storm replay verifies that SWI triggers the HEC-RAS HDF5 ingestion pipeline seamlessly.

The validation results confirm that the system correctly detects hydraulic threats across all 21 corridor sections, with the September 2025 Cévenol storm producing 2 ORANGE sections (speed restriction), 5 YELLOW sections (drainage monitoring), and 14 GREEN sections (all clear). The fragility curve calibration using Tsubaki et al. (2016) field data provides a scientifically grounded safety margin compared to the original uncalibrated parameters.

9. References

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Appendix A: Complete Asset Elevation Database

Table A1: Full asset elevation summary extracted from the 1m LiDAR DTM (107 assets). Assets with zero pixel count indicate that the asset geometry falls outside the DTM raster coverage.

Layer	Asset Label	Elev. Min (m)	Elev. Max (m)	Elev. Mean (m)	Pixels
Voie	Voie_0	204.01	288.66	226.46	16,453
Réseau tiers	Réseau tiers_0	221.73	223.43	222.85	57
Routes	Route_0	209.01	267.36	235.07	6,625
Routes	Route_1	201.11	243.45	216.52	7,872
Talus Terre	Talus Terre_0	205.61	217.02	212.28	1,822
Talus Terre	Talus Terre_3	212.62	237.83	229.12	1,321
Talus Terre	Talus Terre_4	214.82	222.43	218.48	2,498
Talus Terre	Talus Terre_5	220.93	266.06	243.08	8,071
Talus Terre	Talus Terre_11	203.31	206.11	205.05	831
Talus Terre	Talus Terre_12	203.61	214.82	208.21	6,316
Buse	Buse_0	203.21	203.61	203.46	30
Buse	Buse_1	208.32	208.42	208.38	34
Buse	Buse_2	220.03	220.93	220.51	23
Buse	Buse_3	220.02	220.22	220.11	15
Buse	Buse_4	201.71	202.81	202.51	26
Buse	Buse_5	203.71	203.81	203.78	35
Buse	Buse_6	203.61	203.71	203.69	14
Pont Rail	Pont Rail_2	238.05	239.05	238.49	104

Pont Rail	Pont Rail_3	204.71	210.51	207.88	194
Dalot	Dalot_0	213.32	219.62	217.26	165

Appendix B: System Module Inventory

Table B1: Python module-to-layer mapping for the Digital Twin codebase.

Python Module	Layer	Function	Key Inputs	Key Outputs
src/engine/data_ingestion.py	2	Rainfall ingestion	Open-Meteo API / CSV	rainfall_Ligne_400.csv
src/engine/preprocessor.py	2	Data handoff	Raw rainfall	Processed arrays
src/engine/swi_calculator.py	3	SWI + Sigmoid	Rainfall hourly	SWI, C_runoff per step
src/engine/hecras_hdf5_reader.py	3	HDF5 WSE reader	.p02.hdf (422 MB)	WSE per cell per step
src/engine/fragility_curves.py	3	P_failure CDF	Overtopping depth	Risk % per asset
src/engine/alert_dispatcher.py	4	RAMS alerts	Risk %, Z-thresholds	GREEN/YELLOW/ORANGE/RED
src/dashboard/app_main.py	4	Streamlit HMI	All processed data	Interactive dashboard
src/api/main.py	4	FastAPI REST	Query parameters	JSON responses

Document generated: June 2026. All figures and tables reference data from the Tartaiguille corridor digital twin implementation (v0.4.0).